

APPLICATION- BASED QUESTIONS

CO1-ASSESSMENT TOOL-1

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1.Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing,a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Answer:

The Caesar Cipher is a simple and widely known classical encryption technique. It works by replacing each letter in the original message (plaintext) with another letter that is a fixed number of positions ahead in the alphabet. The number of positions is called the shift key. In this example, the shift value is 4.

Plaintext:

MEET AT NOON

Encryption Process

To encrypt the message, each letter is shifted 4 positions forward in the alphabet, while spaces remain unchanged.

- $M \rightarrow Q$
- $E \rightarrow I$
- $E \rightarrow I$
- $T \rightarrow X$
- $A \rightarrow E$
- $T \rightarrow X$
- $N \rightarrow R$
- $O \rightarrow S$
- $O \rightarrow S$

- $N \rightarrow R$

Ciphertext:

QIIX EX RSSR

Decryption Process

An authorized member who knows the encryption rule (shift = 4) can easily retrieve the original message. During decryption, each letter of the ciphertext is shifted 4 positions backward.

- $Q \rightarrow M$
- $I \rightarrow E$
- $I \rightarrow E$
- $X \rightarrow T$
- $E \rightarrow A$
- $X \rightarrow T$
- $R \rightarrow N$
- $S \rightarrow O$
- $S \rightarrow O$
- $R \rightarrow N$

Recovered

Plaintext:

MEET AT NOON

Working of the Caesar Cipher

1. Write the original message (plaintext).
2. Choose a fixed shift value (here, 4).
3. Replace every letter with the letter that is 4 positions ahead in the alphabet.
4. Leave spaces and punctuation unchanged.
5. The resulting text is the encrypted message (ciphertext).
6. To decrypt, shift every letter in the ciphertext 4 positions backward using the same key.

Advantages

- Simple and easy to understand.
- Fast encryption and decryption.
- Useful for learning the basics of cryptography.

Limitations

- Provides very low security.
- Can be easily broken using brute-force or frequency analysis.
- Not suitable for protecting highly confidential information.

Conclusion

Using a Caesar Cipher with a shift of 4, the message "MEET AT NOON" is encrypted as "QIIX EX RSSR". An authorized member who knows the shift key can recover the original message by shifting each letter 4 positions backward. This demonstrates the basic concept of encryption and decryption using a classical substitution cipher.

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption. Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Answer:

A cybersecurity intern intercepts the coded message "GSRH RH Z HVXIVG" during network monitoring. The message appears to use a simple substitution technique instead of a complex encryption algorithm. Based on its letter patterns and structure, the most likely encryption method is the Atbash Cipher, a classical substitution cipher.

The Atbash Cipher replaces every letter in the alphabet with its opposite letter. For example, A is replaced by Z, B by Y, C by X, D by W, E by V, F by U, G by T, H by S, I by R, and so on. This substitution is fixed and does not require a secret key.

Ciphertext:

GSRH RH Z HVXIVG

Decryption Process:

Using the Atbash substitution:

- $G \rightarrow T$
- $S \rightarrow H$
- $R \rightarrow I$
- $H \rightarrow S$

This gives the first word: THIS

- $R \rightarrow I$
- $H \rightarrow S$

This gives the second word: IS

- $Z \rightarrow A$

This gives the third word: A

- $H \rightarrow S$
- $V \rightarrow E$
- $X \rightarrow C$
- $I \rightarrow R$
- $V \rightarrow E$
- $G \rightarrow T$

This gives the fourth word: SECRET

Original

Message:

THIS IS A SECRET

Justification:

The message is most likely encrypted using the Atbash Cipher because it follows a simple substitution pattern where each letter is replaced by its opposite letter in the alphabet. The repeated ciphertext "RH" appears twice and decrypts to "IS", which is a common English word. Similarly, the last word "HVSXIVG" decrypts to "SECRET", forming a meaningful English sentence. The complete decrypted message, "THIS IS A SECRET," is grammatically correct and matches the context of a confidential communication.

Advantages of the Atbash Cipher:

- Easy to understand and implement.
- Does not require a secret key because the substitution rule is fixed.
- Useful for learning the basic concepts of classical cryptography.

Limitations of the Atbash Cipher:

- Provides very weak security.
- Can be easily broken using frequency analysis or by recognizing the substitution pattern.
- Not suitable for protecting confidential information in modern communication systems.

Conclusion:

The intercepted message "GSRH RH Z HVSXIVG" is most likely encrypted using the Atbash Cipher. By replacing each letter with its opposite letter in the alphabet, the original message is

successfully recovered as "THIS IS A SECRET." The repeated letter patterns, meaningful word structure, and fixed substitution rule strongly support this conclusion.

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

Answer:

A company uses a keyword-based encryption technique called the Vigenère Cipher to protect confidential internal communication. Unlike the Caesar Cipher, which uses the same shift for every letter, the Vigenère Cipher uses a keyword to generate different shifts for different letters of the message. This makes the encryption more secure because each letter is encrypted using a different shift value based on the keyword.

In this scenario, the employee wants to send the message "HELLO" using the keyword "KEY".

Plaintext:

HELLO

Keyword:

KEY

Since the keyword is shorter than the plaintext, it is repeated until it matches the length of the message.

Plaintext : H E L L O

Keyword : K E Y K E

Each letter is assigned a numerical value where A = 0, B = 1, C = 2, ..., Z = 25.

Plaintext Values:

H = 7

E = 4

L = 11

L = 11

O = 14

Keyword Values:

K = 10

E = 4

Y = 24

$$K = 10$$

$$E = 4$$

The encryption formula used in the Vigenère Cipher is:

$$\text{Ciphertext} = (\text{Plaintext} + \text{Keyword}) \bmod 26$$

Now encrypt each letter one by one.

Step 1:

$$\text{Plaintext Letter} = H (7)$$

$$\text{Keyword Letter} = K (10)$$

$$7 + 10 = 17$$

17 corresponds to R

Step 2:

$$\text{Plaintext Letter} = E (4)$$

$$\text{Keyword Letter} = E (4)$$

$$4 + 4 = 8$$

8 corresponds to I

Step 3:

$$\text{Plaintext Letter} = L (11)$$

$$\text{Keyword Letter} = Y (24)$$

$$11 + 24 = 35$$

$$35 \bmod 26 = 9$$

9 corresponds to J

Step 4:

$$\text{Plaintext Letter} = L (11)$$

$$\text{Keyword Letter} = K (10)$$

$$11 + 10 = 21$$

21 corresponds to V

Step 5:

$$\text{Plaintext Letter} = O (14)$$

$$\text{Keyword Letter} = E (4)$$

$$14 + 4 = 18$$

18 corresponds to S

Ciphertext:

RIJVS

Step-by-Step Encryption Table

Plaintext : H E L L O

Keyword : K E Y K E

Plaintext Value : 7 4 11 11 14

Keyword Value : 10 4 24 10 4

Addition : 17 8 35 21 18

Modulo 26 : 17 8 9 21 18

Ciphertext : R I J V S

Influence of the Repeating Keyword

The keyword "KEY" is repeated as "KEYKE" to match the length of the plaintext. Each letter of the keyword determines the shift applied to the corresponding plaintext letter.

- The first letter K shifts H by 10 positions to produce R.
- The second letter E shifts E by 4 positions to produce I.
- The third letter Y shifts L by 24 positions to produce J.
- The keyword then repeats with K, shifting the next L to V.
- Finally, E shifts O to S.

Because different letters of the keyword apply different shifts, the same plaintext letter can produce different ciphertext letters depending on its position. This makes the Vigenère Cipher much stronger than the Caesar Cipher, which uses only one fixed shift for the entire message.

Advantages

- More secure than the Caesar Cipher because it uses multiple shift values.
- The repeating keyword creates different ciphertext letters for different positions.
- Makes frequency analysis more difficult.
- Easy to implement and suitable for learning classical cryptography.

Limitations

- If the keyword is short and repeated many times, attackers may discover the keyword using cryptanalysis techniques such as the Kasiski examination.
- Less secure than modern encryption algorithms like AES.
- Requires both the sender and receiver to know the same keyword.

Conclusion

The Vigenère Cipher encrypts the plaintext "HELLO" using the keyword "KEY" by repeating the keyword as "KEYKE" and applying a different shift to each letter. The encryption transforms the original message into the ciphertext "RIJVS". The repeating keyword influences each character by changing the shift value for every position, making the encryption stronger than simple substitution methods like the Caesar Cipher.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Answer:

During a digital forensic audit, investigators suspect that a company logo image may contain hidden confidential information. Although the image appears normal when viewed, it may contain secret data embedded using steganography. Steganography is the process of hiding confidential information inside digital media such as images, audio, or video files without noticeably changing their appearance. Unlike encryption, which hides the content of a message, steganography hides the existence of the message itself.

Detecting Hidden Data in the Image

Investigators can use several forensic techniques to determine whether an image contains hidden information.

1. Visual Inspection

The first step is to examine the image carefully for any unusual characteristics. Although steganographic images usually appear identical to the original, investigators may compare the suspected image with the original company logo to identify any visible differences in color, pixels, or image quality. In most cases, no obvious changes will be visible because steganography is designed to avoid detection.

2. File Size and Metadata Analysis

Investigators can compare the file size and metadata of the suspected image with the original version. If the file size is unusually larger without any noticeable increase in image quality or resolution, it may indicate that hidden data has been embedded. Metadata such as creation date, modification date, software used, and image properties may also provide clues about whether the image has been modified.

3. Statistical Analysis

Steganography often changes the statistical properties of an image. Investigators can analyze pixel values, color distributions, and image histograms to detect irregular patterns. These statistical abnormalities may indicate that hidden information has been inserted into the image.

4. Least Significant Bit (LSB) Analysis

Many image steganography techniques hide information by modifying the Least Significant Bit (LSB) of pixel values. Since changing the least significant bit has little effect on the appearance of the image, it is commonly used for hiding data. Investigators can examine these bits using forensic software to determine whether secret information has been embedded.

5. Steganalysis Tools

Specialized forensic software such as StegExpose, Stegdetect, OpenStego, or StegSecret can automatically analyze images for hidden data. These tools examine image structures, detect suspicious modifications, and sometimes recover the embedded information if possible.

Techniques to Strengthen Hidden Communications

To make hidden communications more secure and resistant to detection or extraction, the following techniques can be used.

1. Encrypt the Secret Data Before Embedding

Before hiding the confidential message inside the image, the message should first be encrypted using a strong encryption algorithm such as AES (Advanced Encryption Standard). Even if investigators successfully extract the hidden data, they will only obtain encrypted text, which cannot be understood without the correct decryption key. This provides an additional layer of security beyond steganography.

2. Use Advanced or Randomized Steganography Techniques

Instead of embedding data sequentially into image pixels, advanced steganography methods use random or key-based pixel selection. Only users who know the secret key can determine where the data is hidden. Techniques such as adaptive steganography or transform-domain methods (such as DCT or DWT based embedding) distribute the hidden information in a way that is much harder to detect using statistical analysis or steganalysis tools. This significantly reduces the chances of the hidden data being discovered or extracted.

Advantages of Secure Steganography

- Hides the existence of confidential communication.
- Maintains the normal appearance of the image.
- Provides additional privacy when combined with encryption.
- Difficult for unauthorized users to detect hidden information.
- Suitable for securely sharing confidential digital information.

Limitations

- Hidden data may still be detected using advanced steganalysis techniques.
- Image compression, resizing, or editing may damage or remove the hidden information.
- If weak embedding methods are used, investigators can extract the secret data more easily.
- Steganography alone does not protect the message unless encryption is also used.

Conclusion

In a digital forensic investigation, investigators can detect hidden information in an image by performing visual inspection, analyzing file size and metadata, examining statistical properties, checking the least significant bits of pixels, and using specialized steganalysis tools. To improve the security of hidden communications, organizations should encrypt the secret message before embedding it and use advanced, key-based steganography techniques that distribute the hidden data randomly. Combining encryption with robust steganography provides stronger protection against both detection and unauthorized extraction of confidential information.

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Answer:

A company's legacy communication system uses a transposition-based encryption method with three levels of arrangement, which is commonly known as the Rail Fence Cipher (3 Rails). In a transposition cipher, the letters of the plaintext are not changed; instead, their positions are rearranged according to a specific pattern. Since the characters remain the same and only their order changes, the encryption provides only basic confidentiality.

The received encrypted message is:

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

- a. Determine the Most Probable Original Message

The ciphertext is likely encrypted using the 3-Rail Rail Fence Cipher. To recover the original message, the zigzag arrangement used during encryption is reconstructed, and the characters are read in their original order.

After reversing the transposition process, the original plaintext is:

Original

Message:

WE ARE DISCOVERED FLEE AT ONCE

This is a well-known example used to demonstrate the Rail Fence Cipher.

The decryption process involves:

1. Determining that the message was arranged using 3 rails.
2. Reconstructing the zigzag pattern followed during encryption.
3. Filling the ciphertext characters into the zigzag positions.
4. Reading the letters row by row to recover the original message.

Thus, the encrypted message "WECRLTEERDSOEFEAOCAIVDEN" is successfully decrypted as:

WE ARE DISCOVERED FLEE AT ONCE

b. Why is this Method Vulnerable in Modern Communication Systems?

Although the Rail Fence Cipher was useful in the early days of cryptography, it is considered highly insecure for modern communication because of several reasons.

1. No Character Substitution

The cipher only rearranges the positions of letters without changing the letters themselves. Since the original characters remain unchanged, attackers can easily identify common words and language patterns after trying a few arrangements.

2. Small Number of Possible Keys

The only secret parameter is the number of rails. Since the number of rails is usually small, an attacker can simply try all possible rail values until the original message is recovered. This makes the cipher vulnerable to brute-force attacks.

3. Vulnerable to Frequency Analysis

Because the actual letters remain unchanged, the frequency distribution of letters is preserved. Attackers can use frequency analysis along with pattern recognition to guess the original message.

4. No Protection Against Modern Cryptanalysis

Modern cryptographic attacks use automated tools that can quickly recognize and break simple transposition ciphers. Even long messages can often be decrypted within seconds using readily available software.

5. Lack of Authentication and Integrity

The Rail Fence Cipher only provides basic confidentiality. It does not verify whether the message has been modified during transmission and does not authenticate the sender. As a result, attackers may alter messages without being detected.

Because of these weaknesses, the Rail Fence Cipher is no longer considered suitable for protecting confidential corporate communication.

c. Practical Enhancement to Improve Confidentiality

A practical way to improve the confidentiality of messages is to replace the Rail Fence Cipher with the Advanced Encryption Standard (AES).

AES is a modern symmetric encryption algorithm that encrypts both the characters and their arrangement using a secret cryptographic key. It is widely used by governments, banks, businesses, and cloud service providers because it offers very high security.

The advantages of using AES include:

- Strong protection against brute-force attacks.
- Secure encryption using 128-bit, 192-bit, or 256-bit keys.
- Fast encryption and decryption for large amounts of data.
- Resistant to modern cryptographic attacks.
- Widely accepted as an international encryption standard.

Even if an attacker intercepts the encrypted message, it is practically impossible to recover the original information without the correct secret key.

Conclusion

The ciphertext "WECRLTEERDSOEFEAOCAIVDEN" is decrypted as "WE ARE DISCOVERED FLEE AT ONCE" by reversing the three-rail Rail Fence Cipher. Although this classical transposition method demonstrates the basic principle of rearranging characters for confidentiality, it provides very weak security in modern communication systems. It is vulnerable to brute-force attacks, frequency analysis, and automated cryptanalysis. Replacing the legacy encryption method with a modern algorithm such as AES provides significantly stronger confidentiality, better resistance to attacks, and secure communication for corporate environments.